

ESAT CAMP MOCK TEST D

Physics

EC-PHY-D

PHYSICS

40 MINUTES

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but **do not open this question paper until you are ready to begin.**

This paper contains **27 multiple-choice questions**. There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all of the questions. Each question is worth one mark.

For each question, choose the one option you consider correct. If you make a mistake, erase thoroughly and try again.

You must complete the paper within the time limit.

You can use the question paper for rough working. Dictionaries and calculators are **NOT permitted**.

Please wait until you are ready before turning this page.

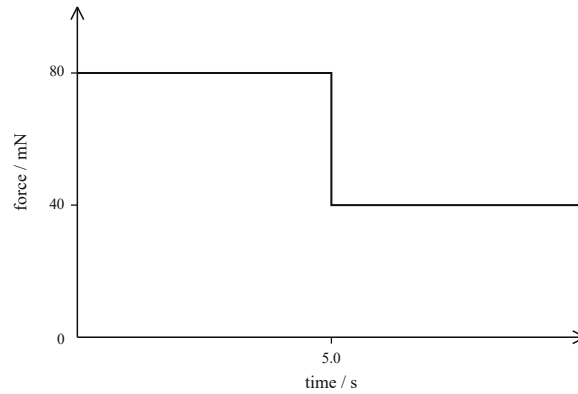
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Physics

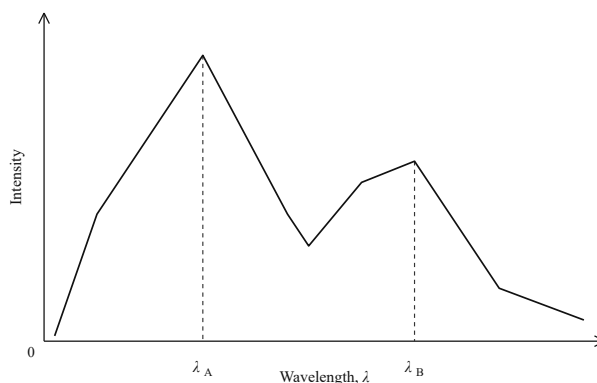
- 1 A small charged sphere is suspended by a thread from a force sensor. The sphere is in a uniform vertical electric field of strength $5.0 \times 10^3 \text{ N C}^{-1}$. The graph shows the reading on the force sensor over time. Initially the reading is constant at 80 mN. At time $t = 5.0 \text{ s}$, the direction of the electric field is reversed, and the reading changes to a new constant value of 40 mN.



What is the magnitude of the charge on the sphere?

- A** 2.0 μC
- B** 4.0 μC
- C** 8.0 μC
- D** 12 μC
- E** 16 μC
- F** 4000 μC

- 2 The figure shows the intensity of an electromagnetic signal as a function of its wavelength, λ .



The signal has two distinct intensity peaks: a taller peak at wavelength λ_A and a shorter peak at a longer wavelength λ_B . This signal is then re-plotted to show its intensity as a function of frequency, f . Let f_A be the frequency corresponding to λ_A and f_B be the frequency corresponding to λ_B . Which statement correctly describes the new graph?

- A The peak originally at λ_A is at a higher frequency than the peak from λ_B , and it is the taller peak.
 - B The peak originally at λ_A is at a lower frequency than the peak from λ_B , and it is the taller peak.
 - C The peak originally at λ_A is at a higher frequency than the peak from λ_B , but it is the shorter peak.
 - D The peak originally at λ_A is at a lower frequency than the peak from λ_B , and it is the shorter peak.
 - E The two peaks will appear at the same frequency on the new graph.
 - F The new frequencies are given by the relations $f_A = c\lambda_A$ and $f_B = c\lambda_B$.
- 3 Two cylindrical rods, X and Y, are used as thermal conductors between two reservoirs maintained at a constant temperature difference. Rod Y has 3 times the thermal conductivity, twice the length, and twice the diameter of rod X.

In the steady state, the rate of heat flow through rod X is P_X and the rate of heat flow through rod Y is P_Y .

What is the value of the ratio $\frac{P_Y}{P_X}$?

- A 2
- B 3
- C 6
- D 12
- E 24

- 4 A fixed mass of an ideal gas is kept at a constant temperature inside a cylinder fitted with a piston. The piston is pulled outwards so that the volume of the gas increases at a constant rate.

Which statement best describes the pressure of the gas?

- A It is constant.
- B It is increasing at a constant rate.
- C It is increasing at a decreasing rate.
- D It is decreasing at a constant rate.
- E It is decreasing at a decreasing rate.

- 5 Two solid spheres, P and Q, fall vertically through a fluid of density ρ . Sphere P has radius r and density 3ρ ; sphere Q has radius $2r$ and density 2ρ .

For this question, the upthrust equals the weight of displaced fluid, and the volume of a sphere is proportional to the cube of its radius. The resistive force on a sphere of actual radius s is $F = ksv$, where v is its speed and k is constant.

Both spheres reach terminal velocities v_P and v_Q . What is v_Q/v_P ?

- A 1
- B 2
- C $\frac{8}{3}$
- D 4
- E 8

- 6 An electric motor has an electrical power input of 160 W. The motor lifts a mass of 8.0 kg vertically at a constant speed of 1.5 m s^{-1} .

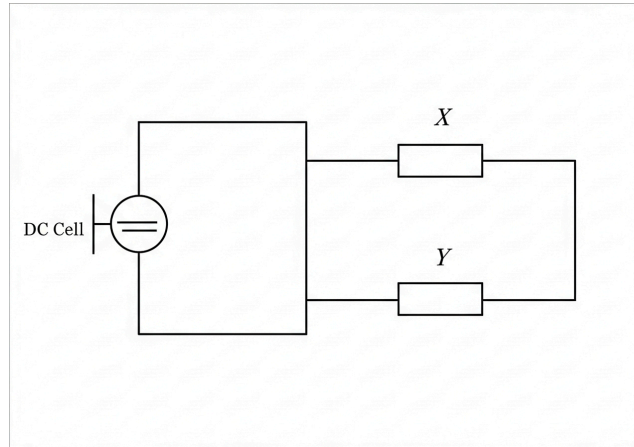
What is the efficiency of the motor? (Assume the acceleration due to gravity $g = 10 \text{ m s}^{-2}$.)

- A 7.5%
- B 15%
- C 25%
- D 40%
- E 50%
- F 75%

- 7 Two wires, X and Y, are made from the same material. Wire Y has twice the length and half the radius of wire X.

The two wires are connected in parallel to a DC power supply.

What is the ratio of the power dissipated by wire Y to the power dissipated by wire X, $\frac{P_Y}{P_X}$?



- A $\frac{1}{8}$
- B $\frac{1}{4}$
- C $\frac{1}{2}$
- D 2
- E 4
- F 8

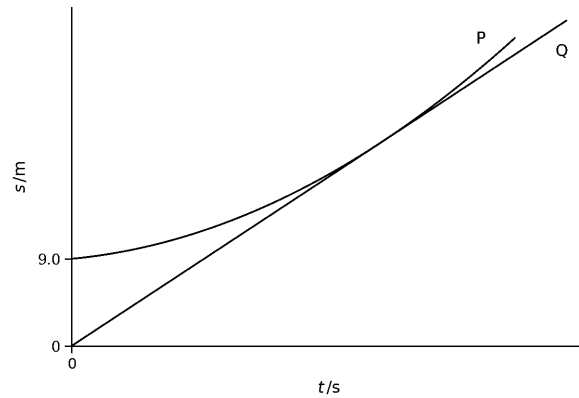
- 8 The displacement–time (s – t) graph shows the motion of two particles, P and Q, travelling in the same direction along a straight line.

Particle P starts at time $t = 0$ from displacement $s = 9.0$ m with an initial velocity of 1.0 m s^{-1} and moves with a constant acceleration of 2.0 m s^{-2} .

Particle Q starts at time $t = 0$ from displacement $s = 0$ and moves with a constant speed v .

On the graph, the straight line representing the motion of Q is tangent to the curve representing the motion of P.

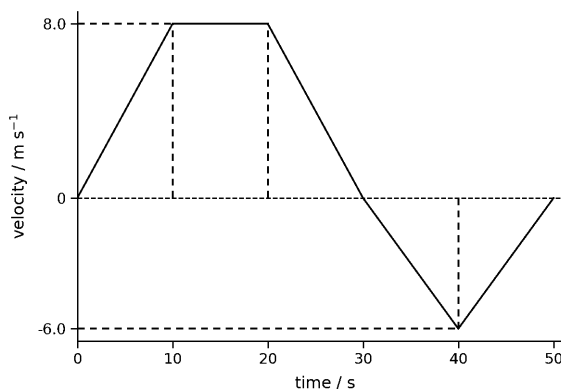
What is the value of v ?



- A 3.0 m s^{-1}
- B 5.0 m s^{-1}
- C 6.0 m s^{-1}
- D 7.0 m s^{-1}
- E 8.0 m s^{-1}
- F 9.0 m s^{-1}
- G 12.0 m s^{-1}
- H 13.0 m s^{-1}

- 9** A small parcel of liquid at the bottom of a container is heated by a source below it. The parcel expands and begins to rise, initiating a convection current. Which statement best explains the physical origin of the net upward force acting on this heated parcel?
- A** The temperature of the parcel increases, giving the molecules sufficient kinetic energy to overcome gravity and move upwards.
 - B** The density of the parcel decreases, causing the weight of the surrounding fluid it displaces to exceed the weight of the parcel itself.
 - C** The volume of the parcel increases, causing the internal pressure of the parcel to become significantly greater than the external pressure from the surrounding fluid.
 - D** The density of the parcel increases, resulting in a stronger reaction force from the container base that pushes the parcel upwards.
 - E** The mass of the parcel decreases due to the expansion, reducing the gravitational force acting on it.
- 10** The graph shows how the velocity of a test vehicle travelling along a straight horizontal track varies with time over a period of 50 s.
- What is the total distance travelled by the vehicle, its distance from its starting position at $t = 50$ s, and the magnitude of its average velocity over the 50 s period?

	Total distance travelled / m	Distance from starting position / m	Magnitude of average velocity / m s^{-1}
A	100	100	2.0
B	100	100	4.4
C	100	220	2.0
D	100	220	4.4
E	220	100	2.0
F	220	100	4.4
G	220	220	2.0
H	220	220	4.4



- 11** An electrical heater with a power output of 100 W is used to melt a large block of ice at 0 °C. The ice melts at a constant rate of 0.20 g s⁻¹.
What is the rate at which thermal energy is lost to the surroundings? (specific latent heat of fusion of ice = 3.5×10^5 J kg⁻¹)

A 30 W
B 40 W
C 70 W
D 100 W
E 170 W
F 70 kW

- 12** A block of mass 4.0 kg rests on a rough plane inclined at an angle of 30° to the horizontal.

The coefficient of static friction between the block and the incline is $\mu = \frac{\sqrt{3}}{6}$.

A light, inextensible string attached to the block passes over a smooth, fixed pulley at the top of the incline. A second block of mass m is suspended vertically from the other end of the string, as shown in the diagram.

What is the complete range of values of m for which the system remains in static equilibrium?

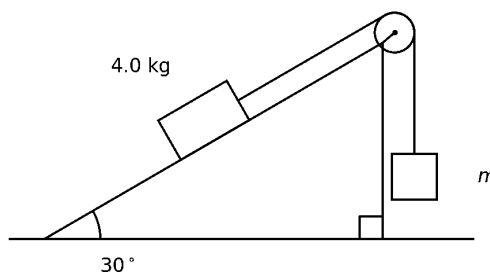


Diagram not to scale

A $1.0 \text{ kg} \leq m \leq 3.0 \text{ kg}$
B $1.0 \text{ kg} \leq m \leq 2.0 \text{ kg}$
C $2.0 \text{ kg} \leq m \leq 3.0 \text{ kg}$
D $1.5 \text{ kg} \leq m \leq 2.5 \text{ kg}$
E $0.5 \text{ kg} \leq m \leq 3.5 \text{ kg}$
F $0.8 \text{ kg} \leq m \leq 3.2 \text{ kg}$

- 13** Two plane mirrors are fixed rigidly together at an angle ϕ . A laser beam travels in the plane perpendicular to the line of intersection of the mirrors. The beam reflects off the first mirror and then the second mirror. The entire mirror assembly is then rotated by an angle θ in the plane of the beam, such that the beam still reflects off both mirrors. What is the angle through which the final emerging beam rotates?

A 0
B θ
C 2θ
D 2ϕ
E 4θ

- 14** A ray of light is directed horizontally towards a plane mirror M_1 which is inclined at an angle of 30° to the horizontal. The reflected ray strikes a second plane mirror M_2 and, after reflecting from M_2 , travels vertically upwards, as shown in the diagram.

Mirror M_1 is now rotated clockwise through an angle α , where $\alpha < 30^\circ$. The direction of the incident ray is unchanged.

In what direction and through what angle should mirror M_2 be rotated in order for the ray to still emerge vertically upwards after reflecting from mirror M_2 ?

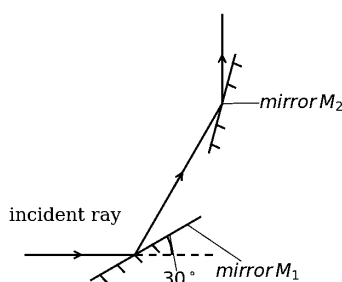


Diagram not to scale

A clockwise through an angle α
B anticlockwise through an angle α
C clockwise through an angle 2α
D anticlockwise through an angle 2α
E clockwise through an angle $\frac{\alpha}{2}$
F anticlockwise through an angle $\frac{\alpha}{2}$

- 15** A ball is projected vertically upwards from the ground. It rises to a maximum height H and then falls back to the ground. Throughout the motion, the ball is subject to a force of air resistance that increases as the speed of the ball increases.

Let W_{up} be the magnitude of the work done by air resistance during the ascent.

Let W_{down} be the magnitude of the work done by air resistance during the descent.

Which of the following statements correctly compares W_{up} and W_{down} ?

- A** $W_{\text{up}} = W_{\text{down}}$, because the distance travelled is the same in both phases.
 - B** $W_{\text{up}} > W_{\text{down}}$, because the speed at any given height is greater during the ascent.
 - C** $W_{\text{up}} < W_{\text{down}}$, because the time taken for the descent is greater than for the ascent.
 - D** $W_{\text{up}} < W_{\text{down}}$, because the ball accelerates due to gravity during the descent.
- 16** A periodic wave travels through a medium with a speed of 5.0 m/s. The period of the wave is 2.0 s.

What is the number of complete wavelengths that fit into a distance of 120 m along the direction of propagation?

- A** 12
 - B** 24
 - C** 48
 - D** 60
- 17** A variable resistor is connected to a power supply that provides a constant voltage. The resistance of the resistor decreases at a constant rate over a period of time, but remains positive.
- Which statement correctly describes the power dissipated by the resistor during this time?
- A** It is increasing at an increasing rate.
 - B** It is increasing at a constant rate.
 - C** It is increasing at a decreasing rate.
 - D** It is not changing.
 - E** It is decreasing at a constant rate.
 - F** It is decreasing.

- 18** A small cylindrical magnet with a uniform cross-sectional area of 20 cm^2 floats vertically in a liquid of density 800 kg m^{-3} . An external magnetic field exerts a vertical force on the magnet.

When the direction of this magnetic field is reversed, the magnitude of the force is unchanged, but its direction is inverted. This causes the total change in the submerged height of the magnet to be 5.0 mm .

What is the magnitude of the magnetic force?

(Assume the gravitational field strength $g = 10 \text{ N kg}^{-1}$.)

- A** 0.0040 N
 - B** 0.040 N
 - C** 0.080 N
 - D** 0.40 N
 - E** 0.80 N
- 19** An electromagnet is placed on a top-pan balance and produces a uniform horizontal magnetic field of strength 0.50 T .
A horizontal beam of protons is fired through the field, perpendicular to the field lines. The experiment is then repeated with a beam of electrons at the same speed, travelling in the same direction.
At any instant, there are 1.0×10^{10} particles within the magnetic field.
The difference between the two balance readings is 1.6 g .

What is the speed of the particles?

(elementary charge, $e = 1.6 \times 10^{-19} \text{ C}$; gravitational field strength, $g = 10 \text{ N kg}^{-1}$)

- A** $1.0 \times 10^5 \text{ m s}^{-1}$
 - B** $2.0 \times 10^7 \text{ m s}^{-1}$
 - C** $1.0 \times 10^7 \text{ m s}^{-1}$
 - D** $4.0 \times 10^7 \text{ m s}^{-1}$
 - E** $1.0 \times 10^{10} \text{ m s}^{-1}$
- 20** The chart shows a grid of nuclides, plotting proton number (Z) against neutron number (N).
Nuclide P is at ($N = 90$, $Z = 60$) and nuclide Q is at ($N = 85$, $Z = 57$).
A nucleus of P decays to a nucleus of Q through a sequence of alpha (α) and beta-minus (β^-) decays only.

Which sequence of emitted particles accounts for this transformation?

- A** 1 α particle and 2 β^- particles
- B** 1 α particle and 3 β^- particles
- C** 2 α particles and 1 β^- particle
- D** 2 α particles and 2 β^- particles
- E** 3 α particles and 3 β^- particles
- F** 4 α particles and 5 β^- particles

- 21** A particle is projected from a point O with speed u . The particle passes through a target point P located at a horizontal distance d and vertical height h from O .
Which of the following expressions gives the minimum value of u required for the particle to reach P ?
- A** $\sqrt{g}$
 - B** $\sqrt{2}$
 - C** $\sqrt{g}$
 - D** $\sqrt{2}$
 - E** $\sqrt{g}$
 - F** $\sqrt{g}$
- 22** A resistor of constant resistance dissipates 100 W when the potential difference across it is V . The potential difference is reduced to $0.8V$.
What is the new power dissipated by the resistor?
- A** 64 ext W
 - B** 80 ext W
 - C** 100 ext W
 - D** 125 ext W
- 23** A progressive wave of frequency 5 Hz travels from point P to point Q . The distance from P to Q is 6 m. An observer at Q detects a wave peak 0.10 s after an observer at P detects a peak.
If the wavelength is known to be between 3 m and 5 m, what is the speed of the wave?
- A** 12 m/s
 - B** 20 m/s
 - C** 30 m/s
 - D** 60 m/s

- 24** Two spherical organisms, X and Y, have the same density and specific heat capacity. The radius of organism Y is three times the radius of organism X ($r_Y = 3r_X$). Both are exposed to a radiant heat source that provides uniform energy per unit surface area.

Which option correctly identifies the ratio of total heat energy absorbed per unit time by Y compared to X, and the ratio of the initial rate of temperature rise of Y compared to X?

- A** Energy absorbed: 3 : 1; Rate of temperature rise: 1 : 3
 - B** Energy absorbed: 9 : 1; Rate of temperature rise: 3 : 1
 - C** Energy absorbed: 9 : 1; Rate of temperature rise: 1 : 3
 - D** Energy absorbed: 27 : 1; Rate of temperature rise: 1 : 1
 - E** Energy absorbed: 27 : 1; Rate of temperature rise: 1 : 9
- 25** A solid cuboid has a volume of 0.0040 m^3 and a uniform density of 2000 kg m^{-3} . The areas of its three distinct faces are in the ratio 1 : 2 : 8.

What is the maximum pressure the cuboid can exert when it rests on a horizontal surface? (Use gravitational field strength $g = 10 \text{ N kg}^{-1}$)

- A** 100 Pa
 - B** 800 Pa
 - C** 1000 Pa
 - D** 4000 Pa
 - E** 8000 Pa
- 26** A sealed, fixed-volume vessel containing an ideal gas is placed in a large room of constant temperature and allowed to cool. The rate of heat loss from the gas is proportional to the temperature difference between the gas and the room. At time $t = 0$, the pressure of the gas is 240 kPa. At $t = 20$ minutes, the pressure is 120 kPa. The half-life of the excess pressure is 10 minutes. The excess pressure is the difference between the gas pressure and its final equilibrium pressure.

What are the final equilibrium pressure of the gas and the pressure at $t = 30$ minutes?

- A** $P_f = 80 \text{ kPa}$; $P(30) = 90 \text{ kPa}$
- B** $P_f = 80 \text{ kPa}$; $P(30) = 120 \text{ kPa}$
- C** $P_f = 120 \text{ kPa}$; $P(30) = 135 \text{ kPa}$
- D** $P_f = 80 \text{ kPa}$; $P(30) = 100 \text{ kPa}$
- E** $P_f = 0 \text{ kPa}$; $P(30) = 60 \text{ kPa}$

- 27** An object of mass 2.0 kg is acted on by a constant horizontal driving force of 10 N and a constant horizontal resistive force F . Its velocity increases uniformly from 2.0 m s^{-1} at $t = 0$ to 8.0 m s^{-1} at $t = 2.0\text{ s}$. What is the magnitude of the resistive force F ?
- A** 4.0 N
 - B** 5.0 N
 - C** 6.0 N
 - D** 8.0 N
 - E** 10 N
 - F** 14 N